

# Week 10: Additional Tools & Resources

## Key Concepts:

- **Cloud Computing Environments:** Moving beyond **GitHub Codespaces** (CPU only) to explore other platforms like **Lightning.ai** (with GPU), and **Posit Cloud** (for RStudio).
- **Hardware Acceleration (GPU vs. CPU):** Understanding that Graphics Processing Units (GPUs) can process massive datasets (like thousands of pages of text) significantly faster than standard computer processors (CPUs).
- **Advanced Natural Language Processing (NLP):** Extracting business insights from unstructured text:
  - **N-grams:** Looking at single words (**Unigrams**) vs. two-word pairings (**Bigrams**) to capture context (e.g., "supply" vs. "supply chain").
  - **Lemmatization vs. Stemming:** Reducing words to their base, dictionary form (lemmatization) rather than just chopping off the ends of words (stemming) to group similar concepts together.
  - **Stop Words & Boilerplate:** Filtering out common grammar words (e.g., "and", "the") and standard legal/corporate jargon (e.g., "forward-looking statements") to reveal true business risks.
- **R vs. Python Ecosystems:**
  - **Python:** A general-purpose language dominant in machine learning, AI, and text analysis.
  - **R:** A language explicitly designed for statistical computing, prized for its concise syntax and deep library of specialized econometrics tools (e.g., Difference-in-Differences analysis).
- **Code File Formats & Version Control:** Understanding the difference between file types:
  - **JSON-based ( .ipynb ):** Standard Jupyter notebooks; great for interactive development but the code isn't easily human-readable behind the scenes.
  - **Plain-text ( .rmd ):** R Markdown files; highly readable, edit-friendly, and easy to version-control or view on a mobile device.
- **Robustness & Sensitivity Analysis:** Strengthening research claims by testing alternative models (e.g., looking at specific timeframes like 2010–2019, or analyzing only large firms) to ensure the original regression results hold true under different conditions.

## Practical Skills & Platforms:

- **Navigating New Cloud GUIs:** Setting up a "Studio" on **Lightning.ai** to utilize GitHub Copilot and free GPU hours, and managing Workspaces in Posit Cloud for R-based statistical projects.
- **Automated Document Extraction:** Searching the SEC's EDGAR database via API to find a company's 10-K report, then using text-pattern rules (Regular Expressions) to automatically "scissor out" specifically the *Item 1A: Risk Factors* section.
- **Evaluating Open-Source Tools:** Learning how to select reliable software packages by checking their organizational backing, update frequency, GitHub stars, and documentation quality.

- **Creating Professional Result Tables:** Moving from raw statistical outputs to polished, publication-ready tables that display Descriptive Statistics, Correlation Matrices, and side-by-side Regression Models.

### Packages Introduced:

- **spaCy (Python):** A heavy-duty, state-of-the-art NLP library that uses AI "Transformer" models to understand English grammar, chunk sentences safely, and lemmatize text. Benefits massively from GPU acceleration.
- **BeautifulSoup (Python):** A web scraping tool used to parse through messy HTML code from SEC filings to extract clean text.
- **WordCloud (Python):** A visual library used to translate word/phrase frequency numbers into intuitive graphics.
- **dplyr & tidyr (R):** Core data manipulation libraries in R used to merge datasets, filter rows, drop missing values ( `drop_na` ), and create new calculated columns ( `mutate` ).
- **stargazer (R):** A powerful formatting package used to generate professional, academic-grade regression tables that display multiple models side-by-side, complete with significance stars (\*\*\*) and standard errors.
- **modelsummary & correlation (R):** Specialized libraries for quickly generating clean summary statistics and correlation matrices.